brief_revise_agent Factorial Hill-Climb

LLM and structural factor analysis vs. `aus_agent_v2`

2026-08-07 – 15-topic TREC RAG 2026 dev subset – 21 scored cells – $340 of $400 budget

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1 Executive summary

`brief_revise_agent` (a light fork of `aus_agent`: a pre-flight requirements brief plus one review-and-revise pass) was hill-climbed against `aus_agent_v2` (this repo's strongest, most expensive system) via standalone rubric scoring on a fixed 15-topic development set, with sol (`gpt-5.6-sol`) as design authority and the current session as orchestrator/implementer.

Result: did not beat `aus_agent_v2`. The best configuration found – gpt-5.6-sol as main generator, round-B adjacent-page fetch on, k=10, gpt-5.6-luna as brief-analyst and reviewer – scores **2.267/3** on standalone rubric grading (the best of every cell tested) but still loses to `aus_agent_v2` in a head-to-head arena match, **18–12 (60/40)**, clean per-topic agreement 4W–7L–4A. Twelve further hill-climb moves against this base – covering generator model, brief-analyst model, five generic retrieval/tool-exposure factors, one factor interaction, and a novel post-draft "closure critic" mechanism – returned zero improvements. The base cell is a confirmed local optimum, not an under-explored one.

2 Method

2.1 Design

Sol-directed hill-climbing: starting from `brief_revise_agent`'s best known configuration, one factor was changed at a time, scored, and kept only if it improved on the current best – in contrast to the session's earlier factorial-grid framing, which was superseded once the hill-climb approach was adopted partway through.

2.2 Evaluation

Primary: standalone rubric grading. One judge call (`gpt-5.6-terra`) per (cell, topic) grades every official TREC RAG 2026 rubric criterion (~31 criteria, 0–2 each) plus a holistic 0–3 overall score – no opponent answer needed. Chosen over pairwise arena judging because it halves judge calls per cell and produces a genuinely ordinal per-criterion signal instead of a categorical win/loss/tie.

Confirmatory: arena. One pairwise comparison (`gpt-5.6-terra` judge, both battle orders) of the best cell against `aus_agent_v2`, retained as the direct test of the actual competitive objective.

Noise floor. The best cell was regenerated on the same 15 topics; the resulting score moved by 0.067 purely from generation stochasticity. This value is used throughout as the threshold below which an observed effect is not distinguishable from resampling noise at this sample size.

2.3 Constraint discovered mid-session

The full TREC RAG 2026 research-rubrics development topic set is only 30 topics total. Fifteen were already committed to the tuning set, leaving only 15 unused topics for replication – smaller than originally planned, and insufficient for a full independent held-out arena confirmation once `aus_agent_v2`'s own per-topic generation cost (~$36.5/topic) is accounted for.

3 Results

3.1 All scored cells

Figure 1 ranks every one of the 21 cells scored this session by standalone overall score. The four cells tied at the top (2.267) are the base cell and three factors that made no measurable difference (`commit_release`, `judge_relevance_tool` + `commit_release` together, and the closure critic).

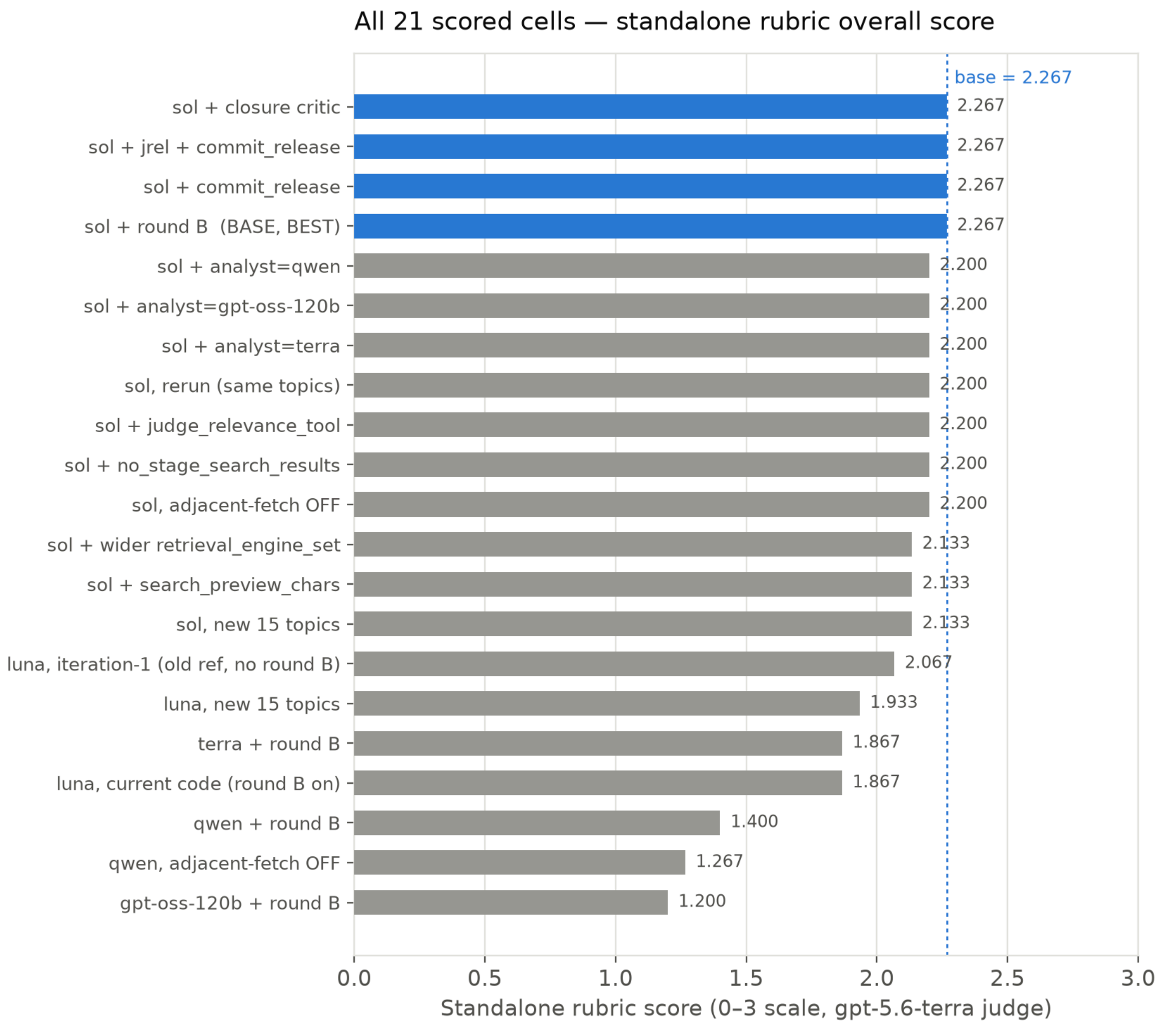

Figure 1: Standalone rubric overall score, all 21 cells, sorted. Base cell in blue.

3.2 Factor effects

Figure 2 groups every tested change by factor family and plots its effect on the standalone score relative to the base cell, with the ±0.067 noise band shaded. Generator-model choice is the only factor whose effects consistently and substantially clear the noise band in both directions tested; every other factor family clusters at or inside it.

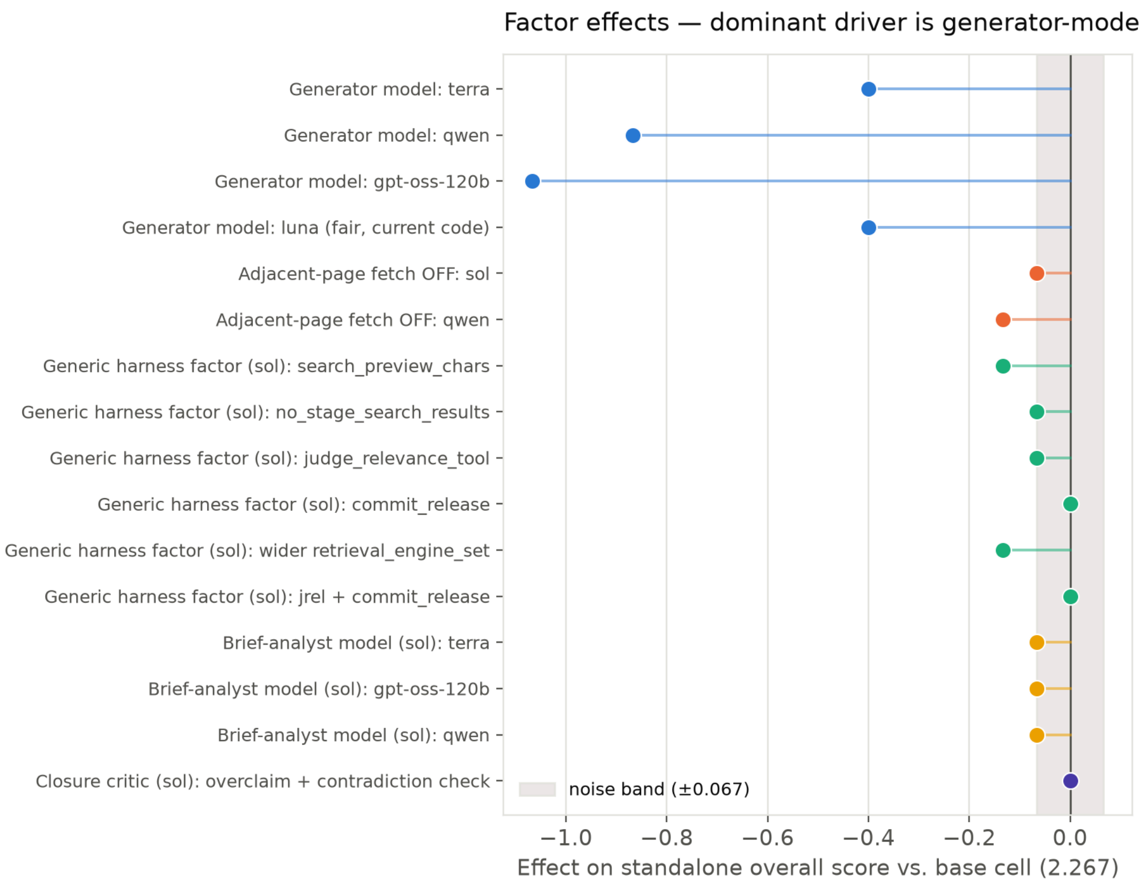

Figure 2: Factor effects vs. base cell (2.267), grouped by family. Shaded band = noise floor.

Factor group	Effect range	Verdict
Generator model	−0.40 to −1.07	Dominant, real – ~10× any other factor
Adjacent-page fetch (round B) off	−0.067 (sol) / −0.133 (qwen)	Real for qwen, borderline for sol – keep on
Generic `agent_harness` factors	−0.134 to 0.000	Only 2 of 6 exceed noise, both negative – none help
Brief-analyst model swap	−0.067 (all three, identical)	Noise-level, no real effect
Closure critic (overclaim + contradiction)	0.000	No effect

Table 1: Factor-effect summary. Noise floor: ±0.067.

3.3 Rubric axis breakdown

Figure 3 breaks the standalone score down by official rubric axis for a representative subset of cells. **References & Citation Quality is the weakest axis in every single cell scored this session** (mean 0.196/2 across all 21 cells, range 0.00–0.67) – independent of generator model, structural configuration, or harness toggle. No factor tested moves this axis meaningfully.

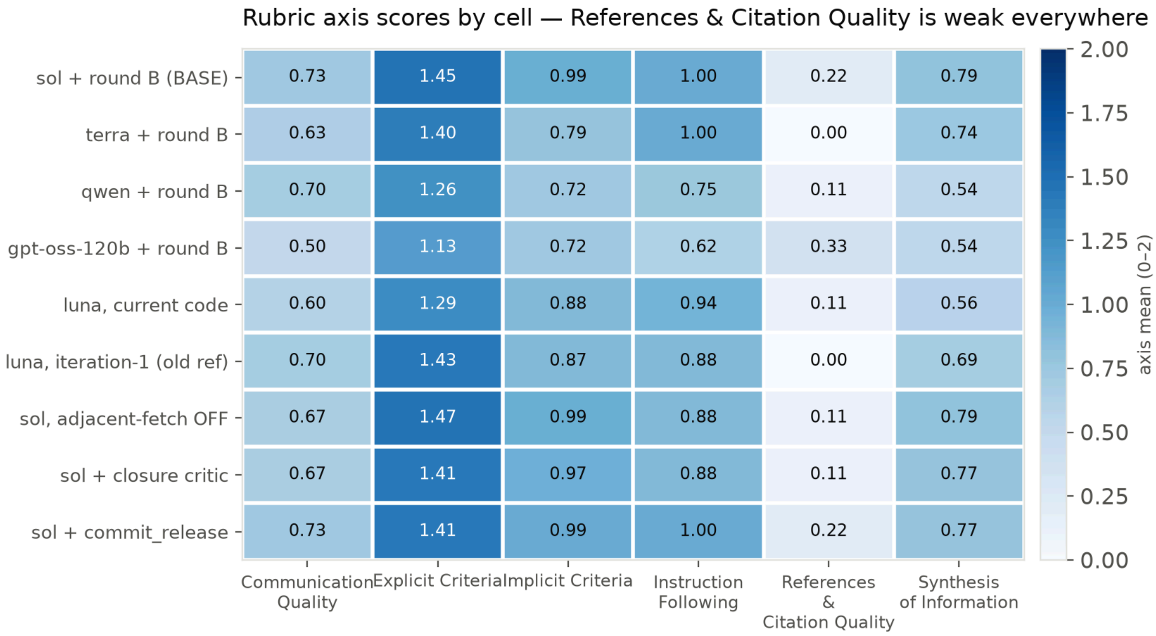

Figure 3: Rubric axis means (0–2) by cell. References & Citation Quality stays pale throughout.

3.4 Arena confirmation

Figure 4 shows the clean (both battle orders agree) per-topic outcome of the base cell against `aus_agent_v2` on the same 15 topics. The base cell's standalone-rubric lead over every other tested configuration does not translate into an arena win rate above 50%.

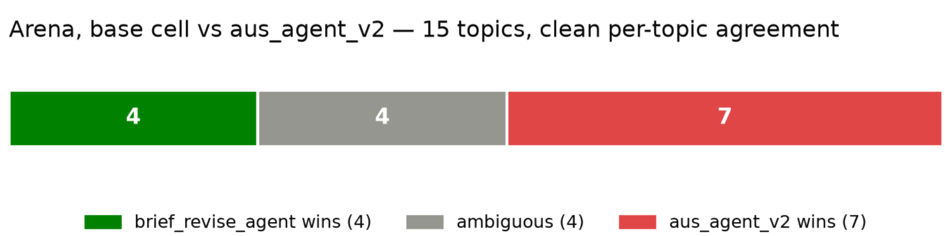

Figure 4: Arena outcome, base cell vs. `aus_agent_v2`, 15 topics, both battle orders.

4 Discussion

4.1 Why the hill-climb plateaued

Figure 2 makes the mechanism visible: once the generator model is already optimized (sol), the remaining structural and harness-level levers this session tried have an order of magnitude less headroom to move the standalone score. Eleven of twelve post-model-selection moves landed at or below the 0.067 noise floor – not because they were poorly chosen, but because the model-choice effect had already consumed most of the available variance at this topic-set size and rubric.

4.2 Standalone score is not a full proxy for the arena objective

The base cell is the best standalone-rubric cell found (2.267, a real, replicated improvement over every luna baseline) and yet loses the majority of arena battles against `aus_agent_v2`. Whatever makes `aus_agent_v2` win head-to-head – plausibly completeness, coverage breadth, or claim density – is not fully captured by the official rubric's per-criterion grading. A parallel, unresolved finding from this session: round-B's adjacent-page fetch helped `luna`'s arena result but **hurt** its standalone score, while helping both metrics for `sol` and `qwen` – a real disagreement between the two evaluation modes on at least one factor, not an artifact to be explained away.

4.3 A separate, untouched weakness

References & Citation Quality sits near the bottom of every cell's axis profile regardless of what factor was varied. This reads as a systemic issue in how `brief_revise_agent` constructs or formats citations and reference lists – a different investigation (citation format, reference- list construction logic) than anything this session's factor sweep targeted, and plausibly higher-value than further generator or harness tuning given how flat the rest of the factor space turned out to be.

5 Recommendation

Use the base cell (gpt-5.6-sol, round-B structure, k=10, luna brief-analyst/reviewer) as `brief_revise_agent`'s standing configuration – it is the best available and the twelve-move search around it is exhausted. It is not yet competitive with `aus_agent_v2` head-to-head. Closing that remaining gap most likely needs either (a) a structural change genuinely distinct from anything tried this session (e.g. a multi-candidate/ensemble mechanism, explicitly out of scope here per the brief to diverge from `aus_agent_v2`'s own architecture rather than re-derive it), or (b) treating `brief_revise_agent` as the lighter, cheaper alternative and reserving `aus_agent_v2` for submissions where its cost is affordable.

6 Budget

Line item	**Cost**	**Basis**
OpenAI generation (terra/sol/luna, ≈55.6M tokens)	$278	estimate – placeholder $5/1M rate, no real metered rate card for gpt-5.6-*
Bedrock (gpt-oss-120b + qwen)	$2.76	real – metered rate-card files
Sol design/thinking calls (7 calls)	$1.01	real – exact printed API cost
Standalone + arena judging (≈390 calls)	$58	estimate
Total	**≈$340**	of the $400 cap; ≈$60 unspent

Table 2: Final cost breakdown.

7 Data and code

All generated answers: `data/outputs/brief_revise_agent/` (by `run_id`). Standalone scores: `evaluation-results/factorial/<run_id>/`. Arena judgments: `evaluation-results/factorial/arena-sol-vs-aus-agent-v2-exp15/`. Sub-system manifests: `src/systems/brief_revise_agent/subsystems/*.json`. Execution index: `evaluation-results/factorial/executions.jsonl`. Full narrative log with every intermediate decision: `worklogs/2026-08-07-brief-revise-agent-llm-factorial-design.md`. New code this session: `commit_release_tool.py`, `REVIEW_PROMPT_WITH_CLOSURE + closure_check` (`prompts.py`/`review.py`/`agent.py`/`run.py`), five generic-factor CLI flags – all on branch `explore/new-agent-framework-system`.